driven swarms. The chapter emphasizes the broad spectrum of drone applications across defense, industry, and society, highlighting their role in surveillance, logistics, agriculture, infrastructure monitoring, and disaster management. Furthermore, it underscores the growing economic and strategic significance of drones in global markets, as well as the integration of autonomy and data-driven decision-making that defines the latest generation of UAVs.

Review Questions

1. What are unmanned aerial vehicles (UAVs), and how do they differ from manned aircraft?
2. Describe the early motivations behind the development of unmanned aerial vehicles.
3. How did military needs drive the initial innovation of UAV systems?
4. In what ways have drones evolved from military to commercial and recreational applications?
5. What technological factors contributed to the rapid growth of drone usage in recent decades?
6. How has the accessibility of drone technology impacted civilian industries?
7. Discuss the societal implications of drone proliferation in public and private sectors.
8. What are the key advantages of UAVs compared to traditional aerial platforms?
9. What are the primary functions of drones in military reconnaissance and surveillance?
10. How have drones changed modern warfare and defense strategy?